

Dynamic Trapping and Printing of Plasmonic Dimers with Optical Vortex Beams

Published as part of *The Journal of Physical Chemistry C* special issue “Naomi Halas and Peter Nordlander Festschrift”.

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Cite This: *J. Phys. Chem. C* 2025, 129, 8262–8269



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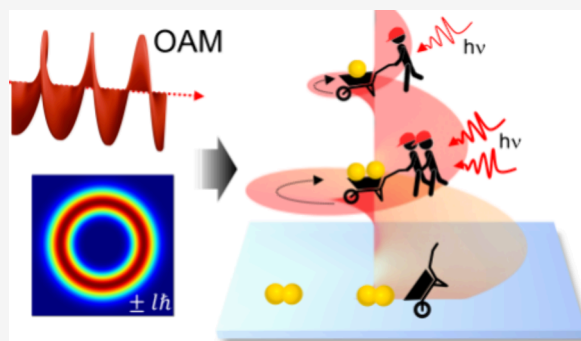


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Supporting Information

ABSTRACT: In this work, we analyze the motion of gold nanospheres in orbital angular momentum (OAM)-carrying optical vortex traps in real time using darkfield microscopy and high-speed video analysis. Notably, we observe that optical binding between gold nanoparticles within the ring-shaped laser trap leads to increased orbiting speeds at a lower focal plane for gold nanoparticle dimers compared to monomers. This behavior is attributed to stronger optical scattering forces acting on the dimers driven by the emergence of a coupled plasmon mode. As the particles move closer together, this mode red-shifts, becoming more resonant with the laser wavelength, eventually causing the system to transition from optical trapping to optical printing. This finding suggests a general mechanism for one-step dimer printing based on plasmonic coupling in vortex beams by adjusting the laser wavelength or modifying the dielectric environment of the nanoparticles via a molecular coating. The feasibility of this approach is demonstrated for optical printing and subsequent surface-enhanced Raman scattering (SERS) spectroscopy on gold nanoparticle dimers coated with 4-nitrothiophenol.



INTRODUCTION

Since the pioneering work of A. Ashkin,¹ optical tweezers have been widely applied in various fields of research, including fundamental physics,^{2,3} cell biology,^{4,5} chemistry,⁶ and nanotechnology.^{7,8} This development has been driven by the continuous search for advanced methodologies to control nanoparticles with light in unprecedented ways. In recent years, the application of structured light, in particular the use of optical vortex beams, has become an area of intense research to unlock new modes and possibilities of particle manipulation.⁹

An optical tweezer operating with a focused Gaussian beam can transfer linear momentum and spin angular momentum (SAM) (when circularly polarized) to a micro-, or nanoobject, which can be used for optical force manipulation¹⁰ and to induce particle spinning in a laser trap.^{11,12} Optical vortex beams (e.g., Laguerre-Gauss (LG)-beams) further carry orbital angular momentum (OAM).¹³ Vortex beams are characterized by a helical wavefront with a phase singularity at the beam center.¹⁴ This gives rise to a doughnut-shaped appearance of a focused vortex beam. Such “doughnut” or ring shaped beams are best known for their application in STED microscopy,¹⁵ but have been used for optical trapping as well.^{16,17}

Optical vortex beams, like Gaussian beams, exert gradient forces and scattering forces.^{18,19} However, in a vortex beam with a doughnut-shaped intensity profile, the gradient force traps particles in the high-intensity region within the ring. The scattering force can be divided into an axial ($\vec{F}_z$) and an azimuthal force ($\vec{F}_\phi$). The axial force arises from transfer of linear momentum and is oriented in the direction of the beam propagation. The azimuthal force arises from transfer of OAM and causes particles to orbit within the ring around the beam axis.^{20,21}

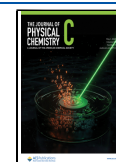
The magnitude of $\vec{F}_\phi$ depends on different parameters, such as the amount of OAM and the optical properties of the particle interacting with the beam. An OAM of $\pm l\hbar$ per photon can be transferred to a particle with an optical vortex. Here, l is the topological charge, i.e. an integer “winding”

Received: February 20, 2025

Revised: March 28, 2025

Accepted: March 31, 2025

Published: April 18, 2025



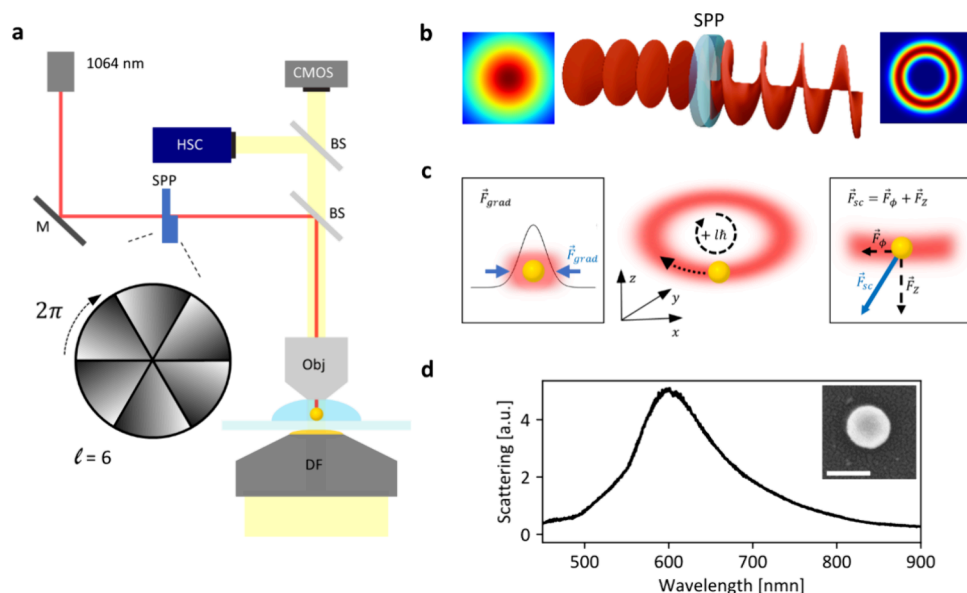


Figure 1. Setup and vortex trap formation. a) Schematic illustration of the DFM setup equipped with an SPP with a topological charge $l = 6$. b) An SPP transforms a Gaussian beam into an optical vortex beam with a helical phase front. The vortex trap displays a doughnut shape in the focal plane. c) Forces acting on a gold nanoparticle in the focal plane of an optical vortex beam. The gradient force $\vec{F}_{\text{grad}}$ trapping a nanoparticle on the ring is shown on the left panel. The scattering force $\vec{F}_{\text{sc}}$ divides into an azimuthal component $\vec{F}_{\phi}$ and an axial component $\vec{F}_z$. The azimuthal force drives the particle movement on the ring circumference due to OAM transfer. The axial force acts in the beam propagation direction. d) Scattering spectra of a single gold nanoparticle. Inset: SEM micrograph of a 100 nm gold nanosphere (scale bar: 100 nm).

number, with its sign indicating the winding direction.¹³ In theory, OAM beams can exhibit infinite values for l .

Furthermore, dynamic light-particle interactions, such as optical binding, can occur when multiple particles are simultaneously trapped by a laser beam.²² The corresponding optical binding forces arise from light-mediated particle-particle interactions and are often referred to as a “third optical force”.²³ For gold nanoparticles, dimerization via optical binding arises from plasmonic coupling.²⁴ This coupling dynamically alters their optical properties since the coupled plasmon mode gradually red-shifts with decreasing interparticle distance,^{25,26} which is qualitatively expressed by the so-called plasmonic ruler equation.²⁷ When the plasmon mode becomes increasingly resonant with the wavelength of the trap, light-particle interactions are enhanced.²⁸ This has major consequences on the balance between gradient and scattering forces acting on coupled particles and further leads to plasmonic heating.

For traditional Gaussian beams, it has been shown that the interplay between optical forces and laser heating opens up a wide toolbox for manipulating plasmonic nanoparticles.^{29–31} The dynamics of light-particle, as well as particle-particle interactions of gold nanospheres with optical vortex beams, however, are only scarcely explored in comparison. In this study, we investigate the potential arising from optical manipulation of gold nanoparticles with helical light beams by demonstrating how the combination of optical forces, plasmon hybridization, as well as plasmonic heating leads to dynamic particle sorting along the beam axis and enables single-step optical printing of gold nanoparticle dumbbells and dimers.

First, we observe that simultaneously trapping several gold nanoparticles in a focused vortex beam leads to dimer formation due to attractive optical binding forces. Notably, these dimers display increased orbiting speeds compared to monomers, while overtaking single particle as they move along

the beam’s circumference. To explain this noninterfering orbital movement of dimers and monomers, we propose an optical sorting process driven by OAM transfer and axial scattering forces, where single spheres orbit closer to the laser focus, while dimers move in lower orbital planes in the direction of the propagating beam. Furthermore, we observe that binding force-mediated dimerization can even induce sufficient plasmonic heating for two separate spheres to amalgamate into dumbbell-shaped particles. These dumbbells exhibit a strongly red-shifted longitudinal plasmon mode, resonant with the trapping laser wavelength, where scattering forces dominate over gradient forces, and optical printing of dumbbells rather than trapping is obtained.

Importantly, this new observation of dumbbell particle formation and printing implies a general strategy for substrate independent, optical printing of closely spaced plasmonic dimers with vortex beams - a feat that can be challenging to achieve by sequential optical printing with a focused Gaussian beam.³² We propose that by either tuning the trapping laser wavelength or by modifying the dielectric environment of the plasmonic nanoparticles, conditions can be obtained, where axial scattering forces move coupled particle dimers toward the substrate, where they are eventually printed. The feasibility of this approach is demonstrated by functionalizing gold nanospheres with 4-nitrothiophenol (NTP) to increase the refractive index of the nanoparticle coating. Indeed, we find that with the NTP coating, dimers featuring subnm gaps can be optically printed in a sequential manner. The presence of a nanogap between the gold spheres forming the dimers is validated through surface enhanced Raman scattering (SERS) measurements of NTP on individual structures. These results demonstrate, that particle manipulation using structured light offers a new and versatile, one-step approach for optical patterning of plasmonic dimers, paving the way for applications in sensing, spectroscopy, nanochemistry, and nanoscale optical manipulation.

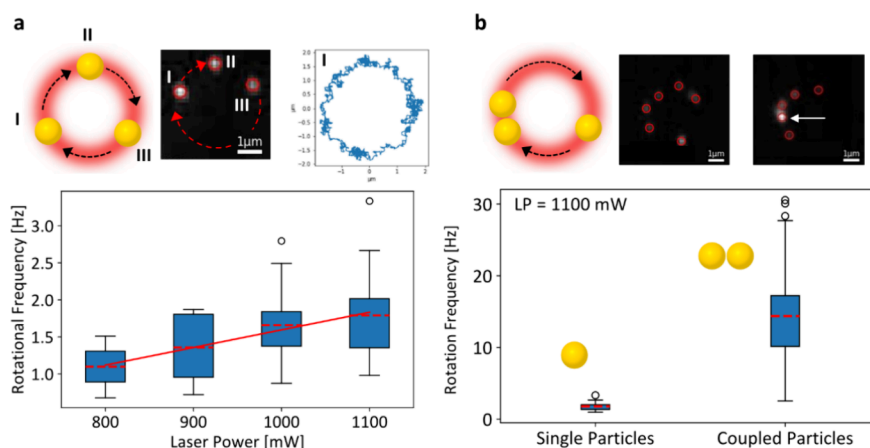


Figure 2. Single particle tracking in an optical vortex trap. a) Top panel: Schematic (left), high-speed camera image (middle), and analyzed trajectory of moving particle I (right). Bottom: Rotation frequencies obtained for single gold nanoparticles at increasing laser powers. b) Top panel: Schematic of dimers and monomers on a doughnut beam (left), high-speed camera images of particles trapped by an optical vortex before (middle) and after dimerization (right). Dimers appear as bright spots in the camera images (a particle dimer is indicated by the white arrow in the right panel). Bottom: Comparison of the rotation frequencies for single particles and dimers obtained with a laser power of 1100 mW. Circles in the boxplot represent outlier data points.

RESULTS AND DISCUSSION

Optical trapping of gold nanospheres with a vortex beam is realized with an upright, dark-field microscope (DFM) setup (Figure 1 a). The microscope is equipped with a CMOS camera (EOS 5D, Canon) to observe single particles entering the laser trap by their scattered light and a high-speed camera (pco dimax HD, pco imaging) to capture videos of the particle motion in real time. A continuous wave (cw), near-infrared (NIR) laser ($\lambda=1064$ nm) is coupled into the microscope objective (63x water objective, numerical aperture (NA) = 1.0) from the top. The optical vortex trap is created by directing the NIR laser beam through a spiral phase plate (SPP) before entering the microscope. This SPP is composed of helical phase steps, which control the phase of the transmitted beam. The topological charge l of the phase plate refers to the number of 2π cycles etched around a 360° turn on the surface (Figure 1b). By increasing the topological charge, the OAM also increases by $\pm l\hbar$, which in turn increases the diameter of the doughnut beam. In our experiments, an SPP with a topological charge of $l = 6$ was determined to provide a suitable balance between the ring diameter ($\sim 3.4 \mu\text{m}$) and the laser power density over the ring area. For a lower l , and consequently a smaller ring diameter, detecting the motion of single particles becomes more challenging with the high-speed camera, as they move along a smaller circumference. On the other hand, as the vortex trap becomes larger, the laser power density on the ring decreases. This reduces the magnitude of the gradient force, requiring higher laser powers to achieve stable trapping. A schematic of the optical forces acting on a gold nanoparticle in a doughnut beam is shown in Figure 1c.

For the experiment, a droplet of water containing citrate-coated gold particles with a diameter of 100 nm is placed on a glass coverslip under the microscope. The citrate coating imparts a negative surface charge to the gold spheres, preventing random particle deposition on the glass substrate through electrostatic repulsion. Before turning the laser on, we diluted the gold nanoparticle solution with deionized water to ensure that only single or a few nanoparticles entered the vortex trap during each measurement.

The particles exhibit a plasmon resonance at ~ 600 nm in water, which is significantly blue-shifted relative to the trapping laser wavelength (Figure 1d). Under these conditions, the gradient force acting on the nanoparticles exceeds the scattering force, enabling optical trapping. However, the presence of scattering forces and OAM transfer is evident, as the particles start orbiting the ring immediately after diffusing into it.

To analyze the influence of laser power on the particle movement, a series of measurements was conducted with laser powers ranging from 800 mW to 1100 mW (measured before the objective) in 100 mW increments (Figure 2). We found that 800 mW was the minimum laser power required to achieve optical trapping. Below this threshold, the gradient force was not strong enough to overcome thermal forces and to keep the particles on the ring. Not only the laser power, but also the size of the gold nanoparticles plays a critical role. Consistent with a previous report by Nakajima et al.,³³ we did not observe trapping of nanoparticles smaller than 80 nm. Again, the gradient force was not strong enough in this case, even for the maximum laser power used. This is explained, since the gradient force scales with the real part of the particle's polarizability α , which for a gold nanoparticle is proportional to r^3 , where r is the particle radius.³⁴

The motion and orbiting velocities of the gold nanoparticles was analyzed using high-speed video microscopy at a frame rate of 6400 fps. Snapshots of three particles orbiting on the ring are shown in Figure 2a. The video frames were analyzed with a single-particle tracking algorithm developed by Crocker and Grier.^{35,36} In brief, particle trajectories were analyzed by extracting the position of each particle for every frame and calculating the temporal change of its angular position.

We found a linear increase in the average rotational frequency from 1.10 to 1.79 Hz as the laser power increased from 800 mW to 1100 mW (Figure 2a). This behavior is expected since the scattering force also increases linearly with the laser power according to $\vec{F}_{\text{Scat}} = \vec{F}_p + \vec{F}_Z = \frac{\sigma_{\text{ext}}}{c} \vec{S}$, with c being the speed of light, $\vec{S}$ the Poynting vector, and σ_{ext} being the particle's extinction cross-section.^{21,37} Further, for an average steady state rotational frequency ($\bar{\omega}$), where particles

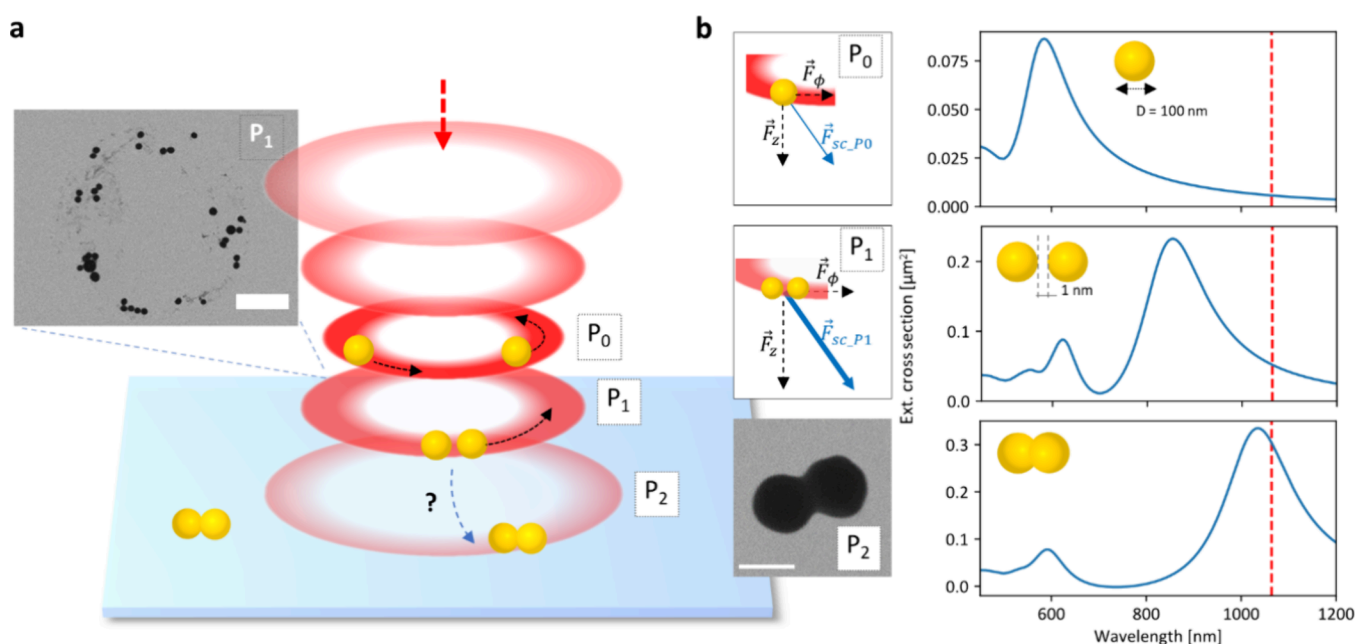


Figure 3. Single and coupled particles move on different orbits. a) Schematic illustration of a loaded vortex beam. The figure inset shows a STEM micrograph measured on a SiN sample (scale bar: 800 nm). Particle deposition in a ring pattern was observed, when the sample was moved into the lower laying orbits in the vortex beam using a piezo stage (from $P_2 \rightarrow P_1$). b) Schematic of the scattering forces acting on a single particle and a dimer at positions P_0 (beam focus) and P_1 (lower orbit). Particle coupling leads to a red-shift of the plasmon coupled mode (FE- simulations, right panel). The stronger scattering force acting on dimers causes them to circulate on a lower orbit in the direction of the beam propagation (indicated by the red arrow in (a)). The red dotted line depicts the wavelength of the laser beam. Dumbbells (STEM micrograph; scale bar: 100 nm) were observed that display a plasmon mode matching the trapping laser wavelength.

are moving with constant velocity, the azimuthal scattering force F_ϕ is equal to the Stoke's drag force according to $F_\phi = F_{drag} = 6\pi\eta r_p R_d \bar{\omega}$, with r_p being the particle radius, R_d being the radius of the vortex ring, and η being the dynamic viscosity of the medium.³³

The rotational frequencies shown in Figure 2 were measured using an objective with an NA of 1.0. Using an objective with a higher or lower NA alters R_d , as well as the field distribution in the focal plane, which in turn also affects F_ϕ and $\bar{\omega}$, respectively.

During the measurement, more and more nanoparticles stochastically diffuse into the vortex trap. Therefore, the likelihood increases that particles interact with each other through optical binding forces.²² In an intense laser field, the simultaneous excitation of two gold nanoparticles results in near-field interactions and coupling of their plasmon modes.³⁸ The optical binding force emerging from dimerization is increasing the closer two particles move together, since the electromagnetic field of a particle plasmon decays exponentially with distance.³⁹ Generally speaking, two nanospheres are pulled closer together due to stronger coupling at short distances, while the coupling strength weakens if the particles drift farther apart. Notably, the optical binding force can be quite strong and exceed the gradient force by a factor of 20.⁴⁰

Dimerization of gold and silver nanoparticles due to optical binding has previously been observed for optical trapping with a focused Gaussian beam.^{41,42} As already mentioned, a distinct feature of plasmon coupling is the emergence of a coupled plasmon mode that gradually red-shifts with smaller particle gaps.²⁶ Since we used an NIR laser for optical trapping, the plasmon mode of a dimer becomes increasingly resonant with the laser wavelength. This results in stronger scattering forces, but also increased plasmonic heating. Both effects can

destabilize the trapping efficiency in a Gaussian beam, but optothermal effects were discussed to be mainly responsible for coupled particles escaping the laser trap.⁴² In a vortex beam, thermal effects of coupled dimers are also expected. However, unlike for a Gaussian beam, where the gradient force pulls both particles toward the center of the beam focus, the particles in a vortex trap can avoid each other by increasing the interparticle distance, while they are still orbiting on the ring. The likelihood of a "thermal escape" due to thermal fluctuations, as observed in Gaussian laser traps, is therefore reduced when using a vortex beam. Consequently, optical vortex beams are advantageous for studying optical binding between plasmonic particles in laser traps.

In the next step, we measured the rotation frequency of coupled particles and compared the result with the data obtained for single particles. The experiment was conducted at a laser power of 1100 mW to increase the trapping efficiency. The movement of particles on the ring was too fast to discriminate between monomers or dimers with the regular CMOS camera or to perform dark-field scattering spectroscopy on individual orbiting particles. However, dimers were identified from high-speed camera images by their relative brightness compared to single particles due to their overall stronger scattering intensity (Figure 2b, upper panel).

An increase in the average rotation frequency by an order of magnitude from 1.79 Hz for monomers to 14.38 Hz for dimers was observed (Figure 2b, lower panel). The higher orbiting velocity of dimers can be attributed to their larger extinction cross-section at the laser wavelength compared to monomers and, connected to that, an increase of the total scattering force. This stronger azimuthal force component $\vec{F}_\phi$ thus induces a larger torque that drives the dimers at higher orbiting speeds.

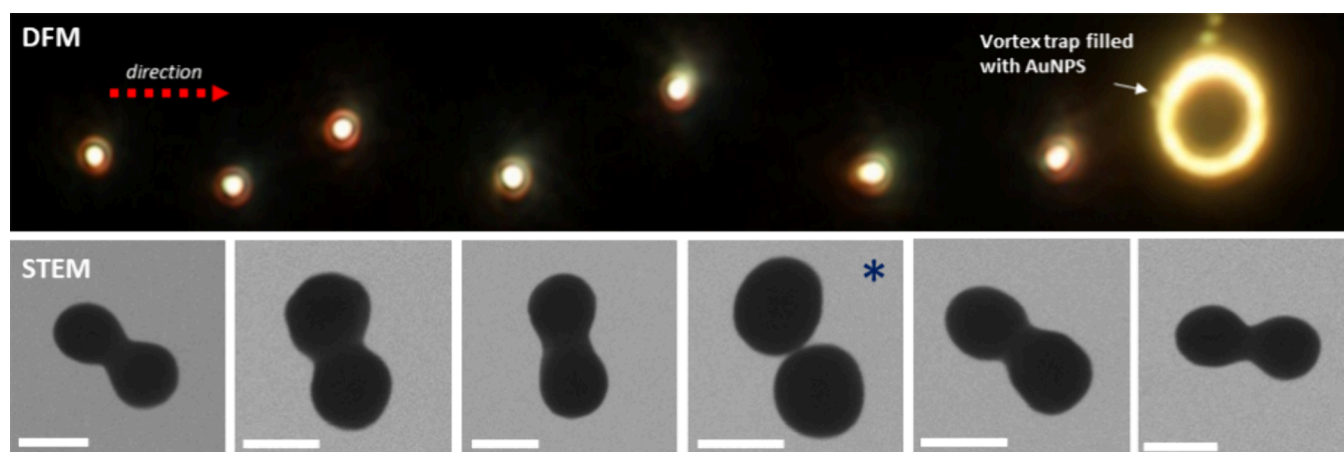


Figure 4. Printing of dumbbell particles and dimers. Top panel: Darkfield image of printed nanostructures and an optical vortex beam loaded with nanoparticles. Bottom panel: STEM micrographs of printed dumbbell shaped nanoparticles (scale bars: 100 nm). An example of a printed dimer separated by a nanogap is indicated by the asterisk.

However, several further factors influence light-particle interactions, contributing both favorably and unfavorably to dimer formation during the measurement. This explains the outliers in the box plot of the dimer data set shown in Figure 2b. As the scattering force increases, so does plasmonic heating. Same as for Gaussian laser traps, thermal fluctuations due to hot-Brownian motion can destabilize dimer formation.⁴³ Additionally, the small heterogeneity in the nanoparticle's size and shape affect the degree of plasmon hybridization, which influences the strength of the optical binding force.⁴¹ Furthermore, the binding force must be sufficiently strong to overcome the electrostatic repulsion between the gold nanoparticles, which is caused by their citrate coating. However, increased plasmonic heating may also cause desorption of the coating, which in turn would then reduce electrostatic repulsion and further promote plasmonic coupling. The delicate interplay of these effects can thus dynamically influence the plasmonic coupling efficiency. For a given laser power, the separation distance of a dimer orbiting along the ring is likely to fluctuate around an average value, where optical binding and thermal forces are at equilibrium. This leads to a broader distribution of the measured rotational frequencies (Figure 2b).

Also uncoupled particles are still moving on the vortex trap. Remarkably, it is observable from high-speed video-microscopy that faster-orbiting dimers appear to overtake these monomers instead of colliding into them. This effect can now be attributed to the axial component of the scattering force. Same as the azimuthal force, the axial force is also enhanced via plasmonic coupling. In a standard optical tweezer with a Gaussian beam, small scattering forces are always present, causing an optically trapped particle to be slightly displaced beyond the beam focus in the direction of the beam propagation.^{44,45} In a similar way, the stronger scattering force acting on faster-moving dimers in a vortex trap also causes them to move at an equilibrium position below the focal plane.

We conducted an experiment to demonstrate that dimers are separated from monomers along the beam axis. A TEM grid with a 50 nm thin silicon nitride (Si_3N_4) window was placed on the microscope stage and gold particles were trapped by the doughnut beam. The TEM substrate was then slowly moved toward the beam focus with a piezo stage. Silicon nitride grids

were chosen to minimize electrostatic repulsion forces between the particles and the substrate. After initial particle deposition was observed on the TEM window, the sample was removed, air-dried and subsequently analyzed using scanning transmission electron microscopy (STEM).

As shown in Figure 3a, we observed that most particles were indeed either paired or arranged in particle chains following a ring-shaped pattern. The primary particle types identified from the STEM measurement include: single spheres, dimers, and dumbbell-shaped particles that appeared to be merged together. Only few single spheres were found on the STEM micrographs, supporting the hypothesis that particles dimerize and cluster due to optical binding, while spatial separation via stronger axial scattering forces pushes coupled nanoparticles into a lower-lying orbit. The presence of dumbbell structures further confirms that plasmonic heating must reach temperatures that are sufficient to fuse two particles together.

The extinction cross sections for single spheres, separated dimers, and dumbbells show that dumbbells exhibit a longitudinal plasmon mode which aligns with the trapping laser wavelength (Figure 3b). Under these conditions, the axial scattering force for dumbbells is expected to dominate over the gradient force, which is the required condition for optical printing. Consequently, dumbbell printing should be experimentally observable.

To explore this idea, we conducted printing experiments in which a vortex trap was loaded with a high number of nanoparticles (>30) simultaneously. When many particles are trapped, a bright yellow ring, formed by circulating particles, is visible during the experiment (Figure 4, top panel). The average number of trapped particles can be determined by counting the particles dispersing from the vortex trap immediately after the laser is turned off (Supporting Information, Figure S1).

Optical printing events are identifiable under the DFM as bright spots appearing on the glass substrate. Following each printing event, the vortex beam was moved to a new location to obtain a sequence of printed particles. The time interval between successive printing steps could not be controlled precisely, since dimerization in the laser trap is a stochastic event, however printing consistently occurred within a few seconds. The printed spots did not align in a straight line but were distributed at different positions along the circumference

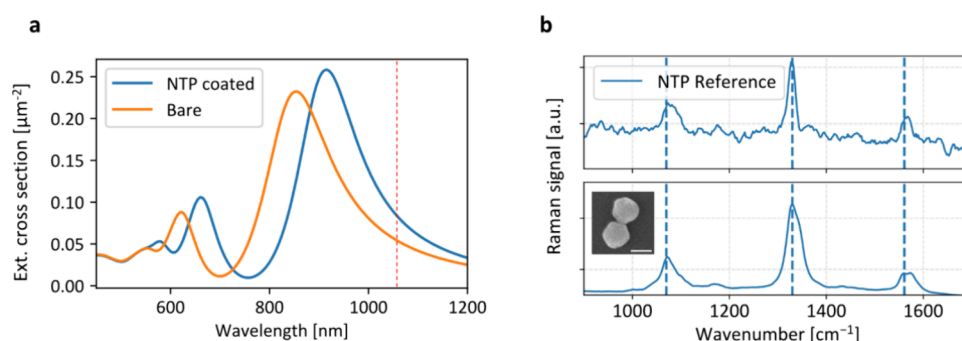


Figure 5. SERS with prefunctionalized printed nanoparticle dimers a) Simulated extinction cross section of a 100 nm gold nanoparticle dimer separated by a 1 nm gap (orange). The extinction cross section of a gold nanoparticle dimer red-shifts by ~ 60 nm when the particles are prefuctionalized by NTP. b) Top: Reference spectrum of NTP conducted on a gold sputtered glass substrate. Bottom: NTP-SERS spectrum obtained from a dimer structure. Inset: STEM micrograph of the measured particle dimer (scale bar: 100 nm). SERS spectra are background subtracted.

of the moving doughnut beam (Figure 4, top panel). STEM measurements revealed that dumbbell particles were obtained from this experiment, with a high yield of $\sim 90\%$. The remaining particles appeared to be closely separated dimers, with a rare occurrence ($< 1\%$) of particle triplets (Supporting Information, Figure S2).

However, this result implies that optical printing of dimers, where an interparticle gap is maintained or even tuned, should be achievable, for instance by using a different laser. In our setup, the trapping laser was resonant with the plasmon mode of a fused dimer, making dumbbell printing the dominant outcome of the process. However, by using a shorter laser wavelength, it should be possible to print separated dimers if their coupled plasmon mode becomes sufficiently resonant with the laser before the particles are molten together.

We could not change the laser in our setup, because the SPP was designed for 1064 nm. Instead of using a different laser, we therefore followed a strategy to induce a red-shift of the particle's plasmon mode by modifying their dielectric environment. This was achieved by prefuctionalizing the gold nanoparticles through a ligand exchange with 4-nitrothiophenol (NTP). Nitrobenzene has a refractive index of ~ 1.53 for 1000 nm at 20°C .⁴⁶ For a gold nanoparticle dimer separated by a 1 nm gap, the NTP coating leads to a shift of the coupled plasmon mode by 60 nm, from 855 nm to 915 nm in water and hence closer to the laser wavelength (Figure 5a). We then repeated the printing experiment with the NTP-functionalized particles. Scanning electron microscopy (SEM) measurements revealed that separated dimers rather than dumbbells were now obtained. This indicates that the scattering force was already sufficient for printing before the particles fused together.

Further, we conducted SERS measurements on printed dimers, to confirm that the particles were indeed separated, and that the field enhancement was not suppressed by electrical contact or electron tunneling.^{47,48} The spectra displayed characteristic SERS peaks of NTP at 1070 cm^{-1} (δ : C–H), 1330 cm^{-1} (ν : NO₂) and 1570 cm^{-1} (ν : C=C), consistent with literature⁴⁹ and with reference measurements of a NTP layer formed on a rough gold film (Figure 5b). Characteristic Raman peaks for citrate were not observed, indicating that citrate coating was completely replaced by NTP, which forms a stable thiol bond with the gold surface. Furthermore, the presence of distinct NTP Raman peaks proves that the temperature reached during the printing

process was insufficient to desorb the molecules from the particle surface. The experiment therefore confirms that dimer printing is feasible using vortex traps when the coupled plasmon mode and the laser wavelength are appropriately aligned. Additionally, our results imply that dimer printing with various interparticle spacings could potentially be achieved by either tuning the laser wavelength or the particle properties, which will be subject to further investigations.

Notably, the concept of nanoparticle printing using vortex beams is not limited to nanospheres and could be extended to other types of particles, such as nanorods and even chiral nanostructures. Through OAM transfer and by varying $\pm \hbar$, it may be possible to control the dynamic photoassembly of nonchiral nanorods into chiral arrangements, similar to what has been obtained by nanofabrication.⁵⁰ Vortex beams may also be useful for studying the photoassisted growth of chiral nanoparticles from nonchiral nanorods, a phenomenon that has already been demonstrated using nonchiral, circularly polarized light.⁵¹

CONCLUSION

We have demonstrated that plasmonic binding forces significantly influence the dynamics of particle motion in an optical vortex trap. Coupled nanoparticles exhibit higher orbiting velocities due to their hybridized plasmon modes being closer to the laser wavelength. This leads to stronger scattering forces acting on dimers, both in the azimuthal direction and along the beam propagation axis. The latter results in a sorting mechanism, where dimers and clusters are separated from single nanoparticles and orbit at a lower equilibrium position beyond the focal plane. Experimentally, this was validated by moving a silicon nitride substrate into the lower particle orbit and analyzing the adhered structures using STEM.

The red-shifted hybridized plasmonic modes of coupled nanoparticles also enable the optical printing of dimers with optical vortex beams. This process further supports the hypothesis of a dynamic sorting mechanism, as the printed structures were predominantly dumbbells. Importantly, this approach allows for controlled outcomes. Whether dimers or fused dumbbells are optically printed can be guided by adjusting the plasmon mode of a coupled dimer relative to the laser wavelength. This concept was experimentally validated through optical printing and SERS measurements using gold nanoparticle dimers functionalized with NTP.

This one-step laser printing method with optical vortex beams provides a rapid and efficient route for patterning functional plasmonic nanodimers with ultranarrow gaps. These structures are particularly well-suited for advanced applications where plasmonic hot-spots displaying high field enhancements are required, such as single molecule SERS and plasmonic catalysis.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.5c01172>.

Experimental details on Materials & Methods; Darkfield micrographs; SEM micrographs (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the Deutsche Forschungsgemeinschaft (DFG) through the Collaborative Research Center SFB1032 (Project no. 201269156, Project A8) and by the ANPCvT (PICT 2019-3458). The authors like to thank Dr. Michael Fedoruk (Vortex Photonics) for helpful discussions. The authors thank local research clusters and centers such as the Center of Nanoscience (CeNS) for providing communicative networking structures. M.A.H. acknowledges CONICET for the external fellowship.

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